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Mesophilic aerobic digestion: An efficient and inexpensive biological pretreatment to improve biogas production from highly-recalcitrant pinewood

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ABSTRACT

Biological pretreatments and their combination with chemical and physical pretreatments are among the inexpensive methods for the improvement of biogas production from lignocellulosic biomass. In this study, mesophilic aerobic digestion was studied as a biological pretreatment to improve biogas production from pinewood. Its effects in combination with other pretreatments, i.e., liquid hot water (LHW), sodium hydroxide (NaOH), and ultrasonic pretreatments, were also investigated. LHW and Alkali (using 8% w/w NaOH) pretreatments were performed at less severe conditions (100 °C for 10 min) to evaluate the synergistic effect of these thermochemical pretreatments on biogas production. The ultrasonic pretreatment was performed at 40 KHz for 30 min at 40 °C. Biological pretreatment was carried out using aerobic sludge at 37 °C for 10 days. The aerobic digestion was the only pretreatment that could individually improve the methane yield from pinewood by 7.3 folds. However, the highest methane yield (57.7 mL/g VS of pretreated pinewood) was obtained by the combination of NaOH, biological, and ultrasonic pretreatments, which was 11.2 folds higher than that of the untreated substrate. It was concluded that the pretreatment with aerobic digestion could significantly improve the biogas production yield, especially in combination with mild thermo-chemical pretreatments.

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1. Introduction

Increase in human activities and the use of unsustainable fossil fuels have caused changes in the climate of the planet [1,2]. This drew attention towards clean and renewable biofuels such as ethanol and biogas that are not only sustainable but can mitigate the climate issues also [3,4]. Biogas, as one of the most environmentally-friendly biofuels, is mainly composed of methane and is produced from the anaerobic digestion of organic materials [5,6].

One of the best sources of organic materials for biogas production is lignocellulosic materials. Lignocellulosic biomass, including forestry, agricultural, and industrial wastes, is an abundant and inexpensive renewable source of organic materials. Nonetheless, in addition to clean and inexpensive fuel production, utilization of these wastes for biofuels production greatly reduces the environmental problems caused due to the accumulation of these wastes [7].

One of the advantages of woody biomass among other lignocelluloses, e.g, agricultural residue such corn stover and wheat straw, is its non-seasonal nature. Woods are divided into two categories: softwoods and hardwoods. Generally, lignin content in softwoods is higher than that of hardwoods. Besides, liginin in softwoods is more complex and difficult to separate. This leads to the requirement of more severe pretreatments on softwoods to increase their digestibility [8].

Although softwood is too difficult to biodegrade due to its highly recalcitrant structure [9], pinewood waste, softwood, is a lignocellulosic compound that can be used as a substrate for biogas production. The main constituents of lignocellulosic materials are cellulose, hemicellulose, and lignin [10]. The strong crystalline structure of cellulose surrounded by lignin and hemicellulose makes a recalcitrant structure that is difficult to be converted to

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biogas by anaerobic bacteria [11]. Therefore, to overcome this problem, it is necessary to pretreat the pinewood prior to biogas production to increase its digestibility [12]. The purpose of pretreatments is to improve the substrate bio-digestibility by disrupting the biomass complex structure as well as by changing the cellulose crystallinity and structure [13,14].

Different types of pretreatments including physical, chemical, physicochemical, biological, and combination of these have been developed over the years [15]. Physical pretreatments, such as grinding and irradiation, usually improve the biomass digestibility through physical breakdown, resizing, crystallinity reduction, increasing the available surface area, and enhancing the biomass pore size [16]. Among physical pretreatments, ultrasonic pretreatment has been commonly used to improve mass transfer in chemical pretreatments. It also forms cavitation bubbles which mechanically disrupt the lignocellulose cell wall [17]. This process can be used to improve biogas production from residual crops, grass, and activated sludge [18]. Ultrasound can be combined with other pretreatments, which provides some advantages, such as the reduction of pretreatment time, temperature, and the amount of chemicals [19]. On the other hand, alkaline pretreatment using sodium hydroxide is one of the widely used thermochemical pretreatments for lignocelluloses. This pretreatment can reduce the lignin content in addition to reduction in cellulose crystallinity. It also increases the biomass porosity and reduces particle size, which result in enhanced digestibility of biomass [20]. Liquid hot water pretreatment is a widely applied physico-chemical pretreatment in which high pressure is applied to maintain water in liquid phase at elevated temperatures [21]. In this process, the mixture of biomass and water is cooked at high temperature and pressure. This type of pretreatment has the potential to increase cellulose degradation and biomass sugar extraction [22,23].

Although most thermochemical pretreatments are very effective in enhancing lignocellulosic biomass biological conversion, they are typically very energy and capital intensive [24]. Biological pretreatments, on the other hand, are less energy intensive and are more environmentally friendly than thermochemical pretreatments [25]. Different bacteria in aerobic sludge (including *Pseudomonadaceae*, *Azotobacter*, and *Neisseriaceae*) were involved in the degradation of lignin and wood structure [26]. Sharma et al. [27] have comprehensively reviewed the biological pretreatments of lignocelluloses. According to this review paper, depending upon the type of microbes used for pretreatment, i.e., monoculture, co-culture, and genetically engineered microbes, biological pretreatments can be categorized into various types. In co-culture pretreatment, finding appropriate microorganisms for co-culture is a challenge [27]. However, aerobic sludge is a microbial consortium that has the ability to live symbiotically, and can be used for biological pretreatment. Thus, in this research, aerobic digestion using the aerobic sludge is proposed as a biological pretreatment to improve the hydrolysis and the digestibility of pinewood.

According to the author's best knowledge, aerobic digestion has not been previously investigated as a biological pretreatment to improve biogas production. The present study aimed at investigating the effect of aerobic digestion of pinewood on biogas production. Due to the highly recalcitrant structure of the softwood pinewood and the low speed of biological pretreatments, the aerobic digestion was also studied in combination with other widely applied pretreatment methods. Ultrasound, sodium hydroxide, and liquid hot water pretreatments were chosen as the widely applied physical, chemical, and physicochemical pretreatments, respectively, to be used prior to the biological pretreatment to find the optimum combined pretreatment method for biogas production from pinewood. The structural changes of pinewood caused by the pretreatments were investigated using compositional analysis, scanning electron microscope (SEM), and Fourier-transform infrared spectroscopy (FTIR).

2. Materials and methods

2.1. Substrate and sludge

The feedstock was prepared from a 10-year-old pinewood tree harvested from the Forest of Isfahan University of Technology. It was dried at 45 °C for 24 h, debarked, and cut into small pieces using a carpentry chopper (Rasha WSH-T64, Iran). The chopped pieces were powdered using a grinder (Bosch MKM6003, Germany), screened using the Mesh 20 sieve, and stored in air-tight plastic bags for further uses. The moisture (or solid) content of the samples was measured by drying them at 105 °C for 6 h.

Aerobic sludge and anaerobic sludge, obtained from the industrial scale bioreactors of the North of Isfahan Wastewater Treatment Plant, working at mesophilic conditions, were used for the biological pretreatment and biogas production, respectively. Aerobic sludge was filtered twice with 20 mesh screen and applied as the inoculum for aerobic digestion. Anaerobic sludge and rumen content were mixed at a weight ratio of 10:1, filtered two times with 20 mesh screen, and stored in dark bottle for 5 d prior to applying as inoculum for anaerobic digestion.

2.2. Pretreatments

The order of the applied pretreatment methods was LHW/NaOH-pretreatment, biological pretreatment, and ultrasonic pretreatment.

2.2.1. Sodium hydroxide (NaOH) pretreatment

For alkali pretreatment, pinewood (5 g) was presoaked in 8% wt. sodium hydroxide solution (95 g) at room temperature for 10 min with stirring. The pretreatment conditions were choosen based on the optimum conditions reported by Salehian et al. [28]. The mixture was then poured in a stainless steel reactor (500 mL, Steel Sanat, Isfahan, Iran) and heated up to 100 °C using an oil bath (Memmert O200, Germany) and hold at this temperature for 10 min. After completion of the pretreatment, the reactor was cooled down in an ice bath, and the mixture was centrifuged at 4500 rpm for 15 min (SELECTALAB, TDZ5-WS, China) at room temperature to separate the solids from liquid phase. The liquid portion was discarded and the pretreated solids were washed repeatedly with room temperature distilled water until the pH of supernatant was close to neutral. Finally, the washed pretreated solids were frozen in liquid nitrogen, freeze-dried (Christ ALPHA 1−2 LD plus, Germany) for 48 h, and stored in plastic bags at 4 °C for further use.

2.2.2. Liquid hot water (LHW) pretreatment

For liquid hot water (LHW) pretreatment, all the steps of alkaline pretreatment (section 2.2.1) were followed using distilled water instead of alkaline solution. The pretreatment conditions were based on the work by Safari et al. [29].

2.2.3. Biological pretreatment

Biological pretreatment was performed at 37 °C and 90 rpm in a shaker incubator (JAL TAJHIZ, Iran). For this, 20 mL aerobic sludge, 5 mL tap water, and untreated or pretreated pinewood (0.25 g dry wt.) were added to the 118 mL open-cap bottles that were transferred to the shaker. The aerobic digestion pretreatment was run for 10 days in order to reach the equilibrium point of total volatile fatty acids (VFAs) production. The VFAs production equilibrium point was determined by performing elementary experiments. These

Table 1
Total carbohydrates (glucan, xylan, galactan, and mannan), lignin, and VS contents of untreated, LHW-, and NaOH-pretreated pinewood solids.

Characteristics	Untreated pinewood	NaOH-pretreated pinewood	LHW-pretreated pinewood
Glucan (%)	40.5 ± 0.2	48.9 ± 1.3	42.0 ± 4.4
Xylan (%)	7.3 ± 0.1	7.8 ± 0.4	4.8 ± 0.2
Galactan (%)	6.4 ± 1.1	5.4 ± 1.8	2.2 ± 0.3
Mannan (%)	7.9 ± 0.2	4.4 ± 3.2	2.6 ± 0.9
AIL[a] (%)	31.4 ± 1.6	24.2 ± 0.6	32.2 ± 1.2
ASL[b] (%)	0.3 ± 0.0	0.1 ± 0.0	0.3 ± 0.0
Lignin (%)	31.7 ± 1.6	24.3 ± 0.6	32.5 ± 1.2
VS[c] (%)	91.5 ± 2.9	95.9 ± 0.1	97.2 ± 1.8
Lignin removal (%)	–	21.0 ± 2.2	2.2 ± 1.6
Hemicellulose removal (%)	–	49.9 ± 1.3	16.9 ± 0.4

[a] Acid insoluble lignin.
[b] Acid soluble lignin.
[c] Volatile solid.

elementary experiments included kinetic studies during 25 days to find the day (10th day) that resulted in the maximum VFA concentration. The whole suspension obtained from biological pretreatment was used for further stages including ultrasonic pretreatment or anaerobic digestion.

2.2.4. Ultrasonic pretreatment

The untreated and pretreated substrates/slurry (0.25 g dry basis) were added to tap water (5 mL) and were incubated in an ultrasonic bath (Soltec SONICA 3200 L ETH S3, Italy, 180 W) at the frequency of 40 KHz for 30 min at 40 °C.

2.3. Biogas production

Untreated and pretreated pinewood solids were subjected to anaerobic digestion for biogas production using batch fermenters (118 mL dark-colored glass bottles) at mesophilic conditions according to the Hansen et al.'s approach [30]. All the anaerobic digestion experiments were carried out in triplicates and the average values are reported. The bottles, used as batch anaerobic fermenters, were sealed by plastic caps and aluminum covers during the anaerobic digestion experiments. In each bottle, 20 mL of anaerobic sludge was added to 0.25 g substrate. Subsequently, deionized water was added to the fermenters to bring the final volume to 25 mL. For biologically pretreated samples, 20 mL of anaerobic sludge was added to the whole slurry obtained from aerobic digestion. A reference fermenter containing anaerobic sludge, water, and pure cellulose was also run in parallel. Two blank fermentations were also run each in duplicate to evaluate the biogas production from aerobic and anaerobic sludge individually.

Anaerobic conditions were created by purging the fermenters with 95% pure nitrogen for 2 min. The bottles were placed in an incubator (JAL TAJHIZ, Iran) at 37 °C and manually shaken every day. The sampling for gas production was conducted every 3 days for the first the 15 days, and then every 5 days for 40 days fermentation. Methane production yield was calculated and reported in mL methane at normal conditions (0 °C, 10^5 pa) per g VS of untreated or pretreated substrate.

2.4. Calculation of gasoline equivalent

To determine the optimal pretreatment combination, the produced energy was calculated as gasoline equivalent from each metric ton of pinewood for each process. For this calculation, lower heating values of the fuels at 25 °C and 1 bar, 32.7 MJ/m^3 for methane and 32.0 MJ/L for gasoline, were considered [31].

2.5. Analytical methods

The amounts of ash, total solids, volatile solids, carbohydrates,

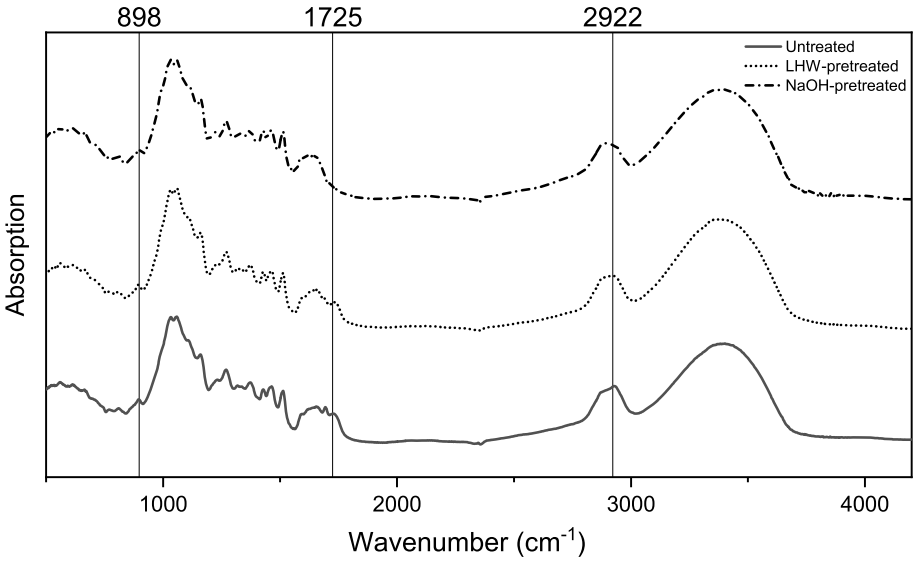


Fig. 1. FTIR spectra of untreated, LHW-pretreated, and NaOH-pretreated pinewood.

Table 2
Characteristics and variation of bands in FTIR spectra of untreated and pretreated pinewood solids.

Wavenumber (cm^{-1})	Functional group	Assignment	Untreated	LHW-pretreated	NaOH-pretreated
3175	Stretching-OH intramolecular hydrogen bonds	Cellulose II	0.29	0.31	0.4
2918	C–H stretching	Cellulose	0.28	0.26	0.32
1730	C=O stretching of acetyl or carboxylic acid	Hemicellulose and lignin	0.17	0.15	0.14
1627	C=C stretching of the aromatic ring	lignin	0.19	0.19	0.27
1598	C=C	lignin	0.17	0.16	0.26
1510	C=C stretching of the aromatic ring	lignin	0.26	0.27	0.37
1465	Asymmetric bending in $C-H_3$	lignin	0.29	0.28	0.38
1430	$C-H_2$ bending	Cellulose	0.27	0.27	0.37
1423	$C-H_2$ symmetric bending	Cellulose I	0.27	0.27	0.38
1375	C–H bending	Cellulose	0.30	0.30	0.36
1335	OH (in-plane bending)	Cellulose	0.29	0.29	0.37
1315	$C-H_2$ wagging	Cellulose	0.29	0.29	0.37
1158	C–O–C asymmetric stretching	Cellulose	0.42	0.44	0.51
1098	β (1–3) polysaccharide weak absorbance	Cellulose	0.51	0.49	0.58
898	Asym., out of phase ring stretching (cellulose)	Cellulose II	0.23	0.22	0.3

and lignin contents in untreated and treated pinewood were determined based on the National Renewable Energy Laboratory (NREL, Denver, Colorado) standard methods [32,33].

High-performance liquid chromatography, equipped with UV/Vis and RI detectors (1260 Infinity, Agilent, California, USA) and Aminex HPX-87 P ion-exchange column (Bio-Rad, Richmond, California, USA), was used to determine the amounts of sugars in the liquid samples. The column temperature was set at 85 °C, and the deionized water was used as an eluent at a flow rate of 0.6 mL/min.

The composition of biogas produced in anaerobic digestion was analyzed by a gas chromatograph (Teif Gostar Faraz Co. GC-2550TG, Iran) equipped with a packed column (Porapak Q, 3 m length and 3 mm internal diameter, Taufkirchen, Germany) and a thermal conductivity detector. The carrier gas helium was used at a flow rate of 20 mL/min. The temperature of the column, injector, and detector was 50 °C, 90 °C, and 140 °C, respectively. The gas samples were injected into the device by a pressure-tight syringe (SGE 250 R-V-GT, Australia). This syringe provides sampling at the reactor pressure. The analytical grade of methane was used as the standard. All the results of the produced methane were reported at the normal conditions (0 °C, 10^5 pa).

The amount of total volatile fatty acids (TVFA) produced after biological pretreatment was calculated according to the Lossie and Pütz's method [27,34] by titrating the substrate with 0.1 N H_2SO_4. In brief, 20 mL of the fermented slurry was measured and placed on a magnetic stirrer to homogenize during titration. In the first step, the titration continued until the pH reached 5, and the amount of acid consumed was noted. After that, the titration continued until the pH reached 4.4, and again the acid consumed was noted. Finally, based on the volume of acid consumed, TVFA was calculated [34]. Also, the amount of each fatty acid produced during the biological pretreatment was determined daily by withdrawing a 2 mL sample, centrifuging, and injecting the supernatant into a gas chromatograph equipped with FID (Beifen Ruili Analytical 197 Instrument Co., Beijing, China) and SolGel-WAX column (30 m length, 0.25 mm internal diameter, 0.25 μm film, SGE, Victoria, Australia).

The structural changes in pinewood caused by pretreatments were investigated using scanning electron microscopy (SEM, XL 30, Eindhoven, Netherlands). The freeze-dried samples were mounted on double-sided tape and placed on an aluminum base. The samples were coated with gold and analyzed by scanning electron microscopy at an accelerating voltage of 15 kV. The crystallinity and structural changes of pinewood after NaOH and LHW treatments were also examined using a Fourier transform infrared (FTIR, WQF-510 A, Rayleigh Analytical Instrument Co., China) spectrometer equipped with a universal ATR (UV-1601, Beifen Ruili Analytical 197 Instrument Co., Beijing, China) accessory and DGTS detector. The FTIR spectra were obtained with an average of 64 scans in the range of 600–4000 cm^{-1} and a resolution of 2 cm^{-1}.

2.6. Statistical analysis

All experiments were carried out in triplicate and the average values were reported. Minitab 19.2.0 software (Minitab Inc., State College, USA) was used for the statistical analysis of the laboratory data using analysis of variance (ANOVA) test and general linear model (GLM). The Tukey method (with $p < 0.05$) was applied to estimate and compare the significant differences among the mean values at 95% confidence level. There are no significant differences (with $p < 0.05$) between the mean values with similar letter groups.

3. Results

3.1. Effect of pretreatments on pinewood composition and structural characteristics

The structural changes of pinewood by LHW and NaOH pretreatments were investigated by compositional, FTIR, and SEM analyses. The amounts of carbohydrates and lignin in untreated and pretreated solids are presented in Table 1. The untreated pinewood was determined to contain 31.7% total lignin and 62.1% carbohydrate, with glucan (40.5%) being the primary carbohydrate. Alkaline pretreatment at 100 °C for 10 min with 8% w/w NaOH mainly decreased the lignin content from 31.7% to 24.2%. However, LHW pretreatment mainly reduced xylan, galactan, and mannan contents by 34.2% (from 7.3% to 4.8%), 65.6% (from 6.4% to 2.2%), and 67.1% (from 7.9% to 2.6%), respectively. As a result of lignin and hemicellulose removal by NaOH and LHW pretreatments, respectively, the corresponding glucan contents were increased up to 48.9 and 42.0% in the solids prepared by these pretreatments.

The effect of NaOH and LHW pretreatments on pinewood structure was also investigated by FTIR analysis (Fig. 1 and Table 2). This analysis is mainly used to determine the changes in the crystalline structure and surface functional groups caused by different pretreatments on lignocelluloses. The peak at 898 cm^{-1}, which is corresponding to the cellulose structure, decreased after pretreatment. This confims partial removal of cellulose due to the pretreatment with LHW and NaOH. Complete removal of the peak at 1725 cm^{-1} was observed after NaOH pretreatment, suggesting the removal of hemicellulose and lignin linkages. The crystallinity index is defined as the ratio of the absorbance at 1098 cm^{-1} to the absorbance at 898 cm^{-1}, which represents cellulose type I and cellulose type II, respectively. The crystallinity index of untreated sample was 2.33, which decreased to 1.93 and 2.11 after NaOH and

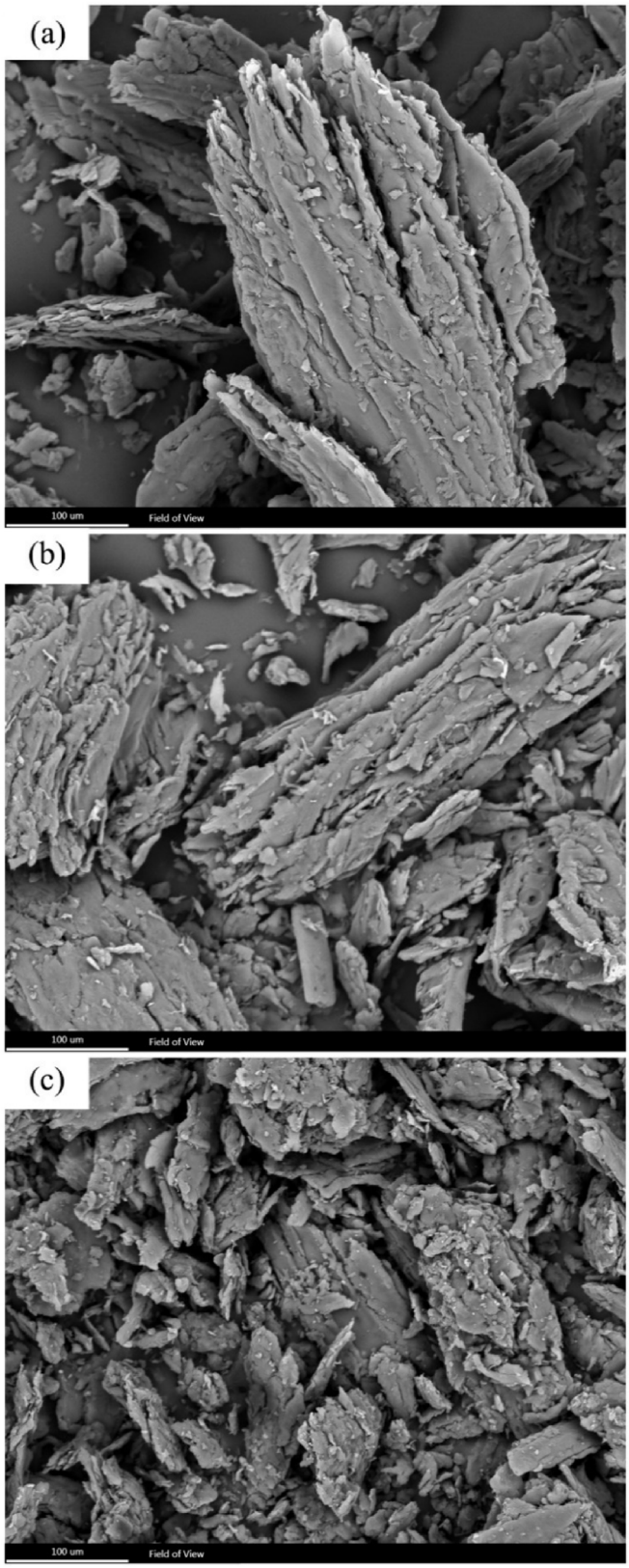


Fig. 2. SEM images of (a) untreated pinewood, (b) LHW-pretreated, and (c) NaOH-pretreated pinewood.

LHW pretreatments, respectively. The total crystallinity index was also calculated from the ratio of absorbance at the wavelength of 1377 cm^{-1} to the absorbance at 2922 cm^{-1}. The total crystallinity index values for untreated, NaOH-pretreated, and LHW-pretreated samples were 1.05, 1.17, and 1.16, respectively.

SEM imaging of the samples was also carried out to investigate the effect of pretreatments on the pinewood structure (Fig. 2). As shown in Fig. 2, LHW pretreatment had a less destructive effect than NaOH pretreatment. NaOH pretreatment degraded the substrate structure and increased its digestibility by removing the major porition of lignin and hemicellulose.

3.2. The effect of biological pretreatment on total volatile fatty acids

The amount of TVFAs produced from aerobic digestion (biological pretreatment) was evaluated and compared with that of anaerobic digestion. The amount of TVFAs produced from the aerobic digestion of untreated wood was 50 mg acetic acid equivalent (HAc_{eq}/L), and it was increased to 91 and 133 mg HAc_{eq}/L applying LHW and NaOH pretreatments, respectively. These values were considerably higher than those of anaerobic digestion (4, 11, and 25 mg HAc_{eq}/L, respectively). The VFAs are intermediate products of anaerobic digestion, which are produced from the acidogenesis stage. The higher content of TVFA produced from aerobic digestion is an evidence of the first two stages of digestion, i.e., hydrolysis and acidogenesis.

The amounts of acetic acid and butyric acid, as the main constitutes of TVFAs, produced during the aerobic digestion of pinewood are presented in Table 3. As the VFA production stopped and the levels of butyric acid and acetic acid took a slight downward trend (at the end of acidification phase) after 10 days, aerobic digestion was stopped. The values of acetic and butyric acids produced from NaOH and LHW pretreated pinewoods were considerably higher than those of untreated samples. The acetic acid concentrations were increased by 82.5% and 166.7% and butyric acid concentrations were enhanced by 83.3% and 250.0% using LHW and NaOH pretreatments, respectively.

3.3. Biogas production

The yields of methane produced during anaerobic digestion of untreated and pretreated pinewood samples were determined and are illustrated in Fig. 3. The highest methane yield of 57.7 mL/g VS of pretreated pinewood was obtained using the combined pretreatments, including sodium hydroxide, aerobic digestion, and ultrasonic pretreatments. The effectiveness of the biological pretreatment (aerobic digestion) in increasing the methane yield (by 7.3 folds) was considerably more than those of sodium hydroxide and liquid hot water pretreatments, and even more than that of ultrasonic pretreatment. At the first look, it seems that all the applied pretreatments, including biological, physical, chemical, and physicochemical pretreatments, improved the methane production yield. However, statistical analysis revealed that the individual effect of NaOH, LHW, and ultrasonic pretreatments was not significant ($P < 0.05$) on the improvement of methane yield. The methane contents in the produced biogas (Table 4) were in the range of 59.1–70.3%. The combination of biological pretreatment with other pretreatments considerably improved the biogas quality in all cases.

3.4. Gasoline equivalent

The results for the equivalent gasoline volume are reported in Table 4. Each metric ton of untreated pinewood produced 1.2 gallons of gasoline. By LHW + BP, 9.8 gallons of gasoline was produced.

Table 3
Types and concentration (g/L) of different volatile fatty acid produced from the biological pretreatment of untreated, LHW-, and NaOH-pretreated pinewood solids.

Time (d)	Untreated pinewood		LHW-pretreated		NaOH-pretreated	
	Acetic acid (g/L)	Butyric acid (g/L)	Acetic acid (g/L)	Butyric acid (g/L)	Acetic acid (g/L)	Butyric acid (g/L)
1	0.43	0.37	0.79	0.68	1.15	0.91
2	0.59	0.27	1.08	0.49	1.58	0.85
3	0.57	0.24	1.06	0.45	1.54	0.78
4	0.63	0.26	1.16	0.49	1.69	0.85
5	0.56	0.21	1.04	0.39	1.51	0.67
6	0.68	0.27	1.25	0.49	1.82	0.85
7	0.74	0.28	1.35	0.51	1.97	0.89
8	0.69	0.27	1.27	0.49	1.85	0.85
9	0.57	0.27	1.05	0.49	1.53	0.85
10	0.63	0.24	1.15	0.44	1.68	0.84

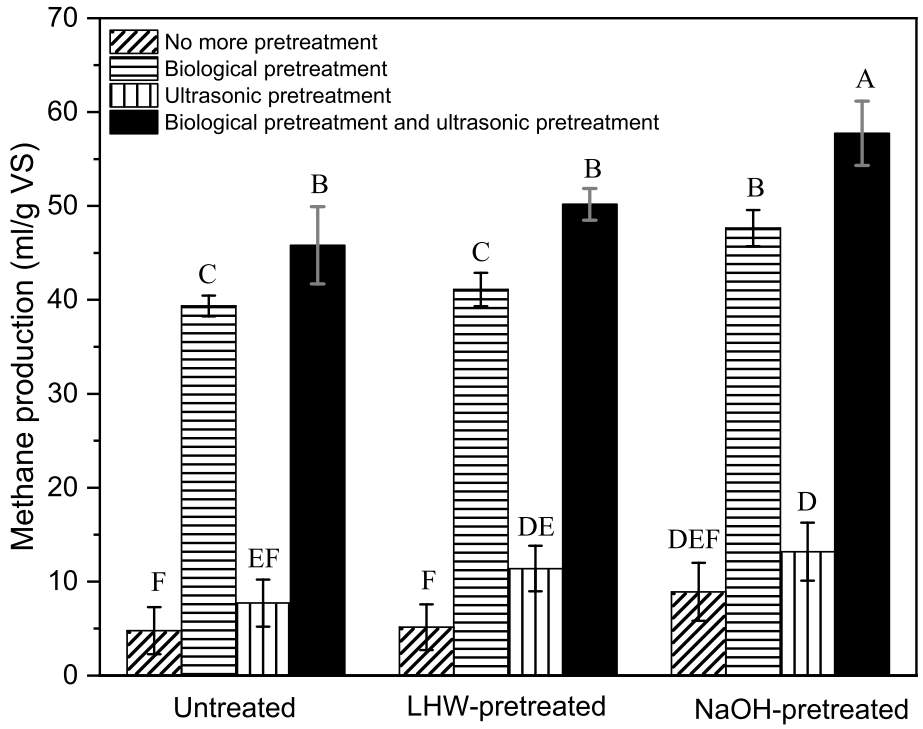


Fig. 3. Accumulated methane yield (mL/g VS of pretreated pinewood) from anaerobic digestion (AD) of untreated, LHW-pretreated, and NaOH-pretreated pinewood applying biological pretreatment and ultrasonic pretreatment. There are no significant differences (with $p < 0.05$) between the mean values with similar letter groups.

Safari et al. [31] report 4.99 gallons of gasoline from pinewood in the optimum pretreatment conditions (1% NaOH, 180 °C, 5 h). Whereas, the present study obtained 2.0 and 2.2 times higher gasoline equivalents using LHW + BP and NaOH + BP pretreatments, respectively.

4. Discussion

The amount of pinewood wastes has increased within the past decades due to its vast use in various industries. The highly recalcitrant structure of pinewood, as a softwood, against microbial conversion promoted this research to find more effective pretreatments.

The NaOH and LHW pretreatments reduced the crystallinity index of cellulose in pinewood by 17.2% and 9.4%, which is an evidence of the partial conversion of type I cellulose to type II cellulose in pinewood. In contrast, the corresponding total crystallinity indexes of pinewood were increased by 11.4% and 10.5%. This can mainly be due to the partial dissolution of lignin and hemicellulose in the NaOH solution and LHW, respectively. Lignin and hemicellulose are amorphous polymers that their removal causes higher crystallinity in the residual lignocellulose solids and, therefore, a higher total crystallinity index. NaOH pretreatment causes the cleavage of ether and ester bindings in the lignin-carbohydrate complexes and also C–C and ester linkages in lignin molecules [35]. These chemical reactions lead to partial lignin removal that was also shown with compositional analysis, which showed 23.3% decrease in lignin content by alkaline pretreatment. LHW pretreatment cleaves the hemiacetal bindings in the lignocellulose structure and causes acids formation [36]. The released acids break the glycoside bindings in the hemicellulose structure, which causes partial hemicellulose dissolution in the solution and decreases the degree of polymerization of hemicellulose [37]. Hemicellulose dissolution by LHW pretreatment was clearly revealed in the composition of pinewood, in terms of decreasing the xylan, galactan and mannan contents by 34.2, 65.6, and 67.1%, respectively. Taherdanak and Zilouei [38] similarly reported that alkali pretreatment of wheat plant at 25 and 75 °C decreased the crystallinity

Table 4
Yields and composition of biogas produced from untreated, liquid hot water pretreated (LHW), sodium hydroxide pretreated (NaOH), biological pretreated (BP), and ultrasonic pretreated (US) pinewood.

Pretreatment applied	CH4 (%)	Biogas yield (mL/g VS of pretreated pinewood)	Methane yield (mL/g untreated pinewood)	Energy (MJ/ton untreated pinewood)	Gasoline equivalents (gallon)
Untreated	61.1 ± 0.7	8 ± 2	4.6 ± 0.0	150.4 ± 0.0	1.2 ± 0.0
BP	62.5 ± 0.8	63 ± 1	37.4 ± 0.3	1223.0 ± 11.5	10.0 ± 0.1
US	61.4 ± 0.7	13 ± 2	7.6 ± 0.1	248.5 ± 3.8	2.0 ± 0.0
BP + US	63.0 ± 0.9	73 ± 4	44.1 ± 0.4	1442.1 ± 15.3	11.9 ± 0.1
LHW	61.5 ± 0.5	8 ± 2	4.3 ± 0.0	140.6 ± 0.0	1.1 ± 0.0
LHW + BP	66.2 ± 0.6	62 ± 2	36.2 ± 0.2	1183.7 ± 7.7	9.8 ± 0.1
LHW + US	61.6 ± 0.3	18 ± 2	9.8 ± 0.0	320.5 ± 0.0	2.6 ± 0.0
LHW + BP + US	67.0 ± 0.5	75 ± 2	44.4 ± 0.2	1451.9 ± 7.7	12.0 ± 0.1
NaOH	59.6 ± 0.7	15 ± 3	7.8 ± 0.0	255.1 ± 0.0	2.1 ± 0.0
NaOH + BP	69.9 ± 0.6	68 ± 2	41.5 ± 0.3	1357.0 ± 11.5	11.2 ± 0.1
NaOH + US	59.1 ± 0.9	22 ± 3	11.4 ± 0.1	372.8 ± 3.8	3.1 ± 0.0
NaOH + BP + US	70.3 ± 0.6	82 ± 3	50.4 ± 0.3	1648.3 ± 11.5	13.6 ± 0.1

BP- biological pretreatment; US- ultrasonic; LHW- liquid hot water; NaOH- sodium hydroxide.

index by 8.82 and 13.23%, respectively. Salehian et al. [28] also showed that pretreatment of pinewood with 8.0% NaOH solution for 10 min at 0 and 100 °C decreased the crystallinity index of the substrate by 24.8 and 28.6%, respectively. They reported that the corresponding total crystallinity indexes were decreased by 14.91 and 7.74%, which is in contrast with the present research.

The SEM images indicated that the alkaline pretreatment caused more severe decomposition in pinewood structure compared to the LHW pretreatment (Fig. 2). This was also confirmed by the FTIR (Table 2). The strong binding of Na^+ and OH^- with water molecules facilitates breaking down the hydrogen bonds within the wood cellulosic structure and converting type I cellulose to type II cellulose [39].

As shown in Fig. 3, except for the biological pretreatment, none of the individual pretreatments could significantly improve methane yield, although the combination of pretreatments was successful options. This was expected since mild conditions (100 °C, 10 min) were chosen for those pretreatments that are not individually appropriate options for the high recalcitrance structure of pinewood. On the other hand, severe pretreatment conditions are challenging for anaerobic digestion and can decrease methane yield [38,39] due to inhibitors formation, total removal of hemicellulose, and too high availability of organic materials that lead to overloading. Therefore, the combination of several mild pretreatments can overcome these challenges.

The combination of NaOH, aerobic digestion, and ultrasonic pretreatments resulted in the highest methane yield, which was 11.1 folds higher than that of untreated one. The NaOH pretreatment removes lignin, increases porosity, and changes the chemical structure of pinewood [40]. These modifications increase the biomass porosity and access to carbohydrates for aerobic bacteria, which hydrolyze the carbohydrates to smaller molecules. After that, sonication makes further structural changes in the lignocelluloses and also causes the disintegration of aerobic sludge by making bubbles and cavitation. The highest methane yields were obtained from biological and ultrasonic pretreatment of LHW-pretreated pinewood < biological and ultrasonic pretreatment of pinewood < biological pretreatment of NaOH-pretreated pinewood. These three combined treatments increased methane production yields by 8.99, 8.60, 9.52 folds, with no significant differences. Chandra et al. [41] reported 111.6% increase in methane produced from alkali pretreated wheat straw (4% NaOH, 37 °C for 120 h), which is much lower than the increase by the combined pretreatments in the present work.

The impact of aerobic digestion pretreatment (Table 3) on methane production was the highest among the other pretreatment methods applied. In fact, the biological pretreatment can be considered as a preliminary digestion stage, in which hydrolysis and acidogenesis steps are performed at aerobic conditions prior to the main anaerobic digestion stage. Hydrolysis and acidogenesis stages are performed by facultative and obligatorily anaerobic bacteria. The involved microorganisms in this preliminary aerobic digestion stage (biological pretreatment) are the facultative anaerobic bacteria, which have a higher growth rate compared to the obligatorily anaerobic ones [42]. In other words, this process can be considered as two-stage digestion that hydrolysis and acidogenesis phases were aerobically performed in the first stage, and acetogenesis and methanogenesis phases were anaerobically performed in the second stage. This two-stage aerobic-anaerobic digestion process was expected to have higher reaction rates compared to the conventional anaerobic digestion processes, which was confirmed by the results (Fig. 4). The intermediate products of acidogenesis phase are VFAs. The high concentration of TVFAs after aerobic digestion confirmed that the hydrolysis and acidogenesis steps were the main bioreactions in this pretreatment. To evaluate the efficiency of the proposed pretreatment in more details, techno-economic analysis and life cycle assessment are suggetsed for the future work.

Ultrasonic pretreatment besides other pretreatments, especially biological pretreatment, synergistically improved methane yield from pinewood. However, its individual effect was not significant. This improvement is because of the physical breakdown of the lignocellulose structure after the bubbles exploitation, which consequently increases the conversion rate of complex organic compounds into simple degradable compounds. Furthermore, the implementation of these waves after aerobic digestion destroys the aerobic sludge, decreases the particle size and, also intensifies the reduction of the volatile sludge solids. These changes cause a significant increase in sludge biodegradability during anaerobic digestion and increasing biogas production [43,44]. Another involved mechanism is the co-digestion of pinewood and aerobic sludge, which can improve the methane yield by adjusting C/N ratio. Pansripong et al. [45] showed that using ultrasonic waves at frequencies of 37 and 102 kHz increased methane produced from rice straw by 21.8% and 18.75%, respectively. This increase in methane production was reported due to the removal of hemicellulose during the ultrasonic pretreatment.

The efficiency of methane production depends on various factors such as type of pretreatment, pretreatment conditions, anaerobic digestion temperature conditions, and substrate type. The recently published studies on different pretreatments were compared with the results of the present study (Table 5). For

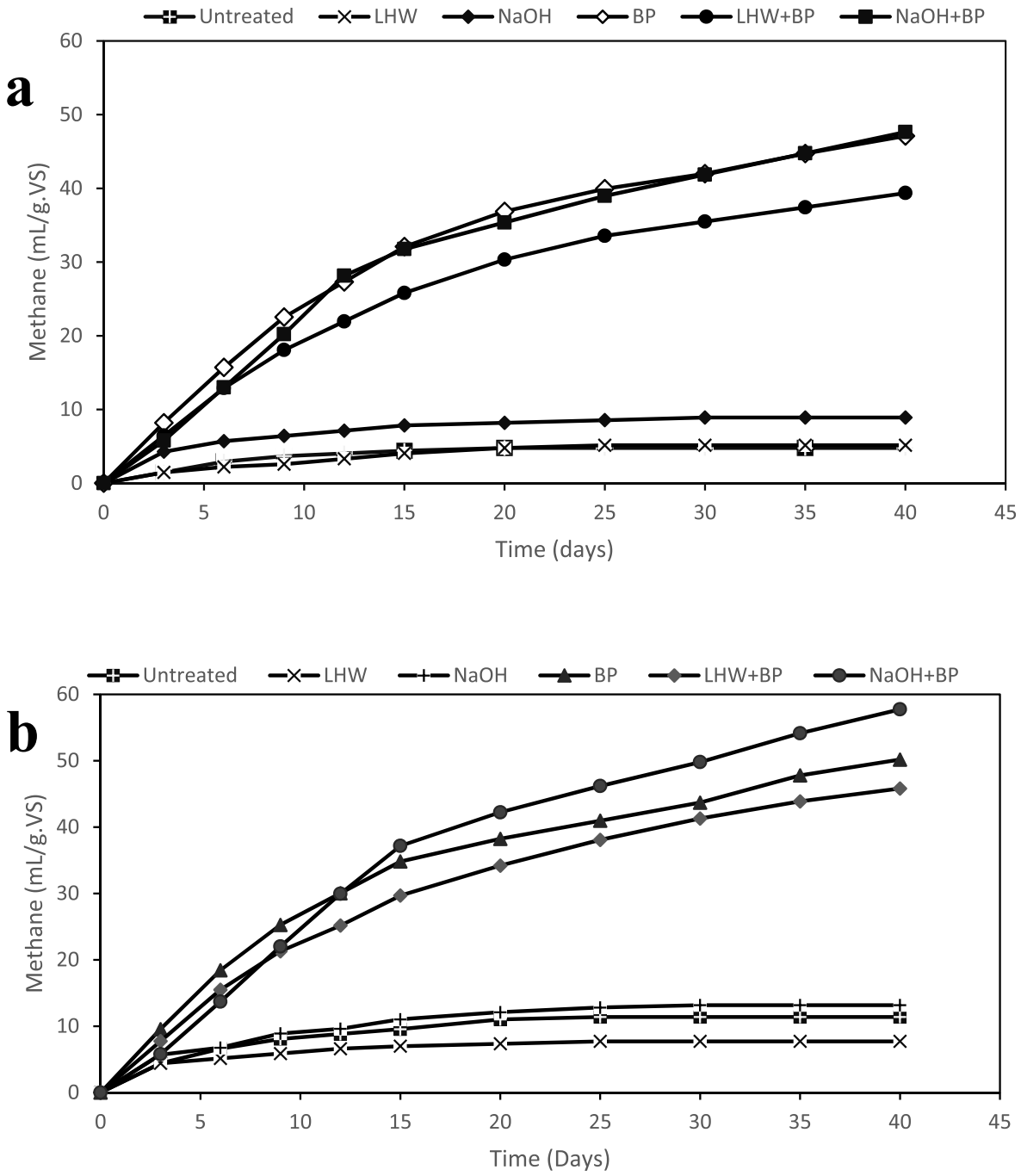


Fig. 4. Time profile of methane production from AD of untreated, LHW-pretreated, and NaOH-pretreated pinewood (a) using biological pretreatment and (b) using biological and ultrasonic pretreatments.

instance, Safari et al. [31] obtained the methane yield of 244 mL/g VS from pinewood by alkaline pretreatment (1% NaOH, 180 °C, 1 h). The higher methane yield, compared to the present study, is due to the application of the liquid fraction of alkaline pretreatment for anaerobic digestion. The liquid fraction of pretreatment has considerable amounts of soluble sugars and a very low volatile solids amount, which is counted in the denominator of biomethane potential. Consequently, these characteristics of the liquid fraction yield a high biomethane potential. Salehian et al. [28] were able to increase the methane production (up to 181.2%) from pinewood using the alkali pretreatment with 8% NaOH at 100 °C for 10 min. However, the increase in methane yields obtained in the present study was unprecedented among the published papers on pinewood, and even on lignocelluloses. However, the final methane yields obtained in this research were within a low range. The observed low range can be corresponded to the growth conditions of the applied pinewood in this research. In other words, many parameters affect the digestibility of lignocellulose from the same species, such as the weather, soil quality, irrigation, and harvest season.

Taherdnak and Zilouei [38] improved the methane yield from wheat plant by 40.8% using alkaline pretreatment (8%NaOH, 75 °C, 60 min). The high methane yield (261 mL/g VS) reported was related to the low amount of lignin and the high amount of starch in

Table 5
Survey of recent studies on methane production from lignocellulosic biomass.

Raw material	Type of pretreatment	Conditions	Anaerobic digestion	Methane production from untreated substrate (mL/g VS)	Increase in methane yield	Reference
Pinewood	Combined pretreatment	8 wt% NaOH, 100 °C, 10 min + Aerobic digestion + Ultrasonic, 40 °C, 30 min	Mesophilic	4.7	1120%	This study
Wheat plant	Alkali	8% NaOH, 50 °C, 60 min	Mesophilic	261	40.8%	[38]
Pinewood	Alkali	8% NaOH, 100 °C, 10 min	Mesophilic	65	181%	[28]
Wheat straw	Alkali	4% NaOH, 37 °C, 120 h	Mesophilic	78.4	87.5%	[41]
Rice straw	Ultrasound	37 KHz, 200 W	Mesophilic	225.2	11.1%	[45]
Pinewood	Organosolv	75% wt. Ethanol, 150 °C, 30 min	Mesophilic	38.7	84%	[46]
Rice straw	Alkali	1.6% NaOH, 30 °C, 24 h	Mesophilic	259	21.4%	[47]
Spruce	Alkali	7% NaOH, 100 °C, 120 min	Thermophilic	30	74%	[48]
Giant reed	Alkali	2% NaOH, 24 °C, 120 h	Mesophilic	217	64%	[49]
Agricultural residue	Biological	Yeast + cellulolytic bacteria + heated sludge	Mesophilic	–	25–96%	[50]
Rice straw	Alkali + LHW + Ultrasonication	3% NaOH, 121 °C, 60 min	Mesophilic	–	150%	[51]
Yard waste	LHW + Microwave	MW 140°C-100 S	Mesophilic	328.9	10%	[52]
Cattle manure + Wheat straw	Ultrasound	52.5 °C, 25.95 S	Mesophilic	102	4.6%	[53]
Corn straw	Biological	30 °C,14 days	Mesophilic	–	131.7%	[54]
Wheat straw	Fungal + Bioaugmentation	*P.ostreatus* and *B.subtilis*	Mesophilic	437	46%	[55]
Cow manure and cereal crop	Fungal	*Trametes Versicolor* aerobic condition	Mesophilic	–	10–18%	[56]

their applied substrate, i.e., wheat plant. Wheras, pinewood, applied in the present work, is a more recalcitrant substrate that wheat straw due to higher lignin content and the stronger ester bonds between cellulose, hemicellulose, and lignin.

5. Conclusions

The aerobic digestion applied as a pretreatment considerably improved methane production yield from the softwood waste. The combination of this biological pretreatment with mild chemical and physical pretreatments synergistically increased biogas production up to 1120%, from 4.6 mL/g VS for untreated pinewood to 50.4 mL/g VS. The biological pretreatment enhances the methane yields by not only disrupting the biomass structure but the aerobic sludge produced during the process also plays a role as a co-substrate that can improve biogas yield by co-digestion mechanism. The high concentration of total volatile fatty acids after aerobic digestion revealed that the biological pretreatment prior to anaerobic digestion was, in fact, a two-stage digestion process, in which hydrolysis and acidogenesis stages were performed in the first stage, and acetogenesis and methanogenesis stages were performed in the second stage. Because of the higher rates of the aerobic bioreactions than that of anaerobic configuration, this proposed two-stage aerobic-anaerobic digestion should have a higher degradation rate compared to the conventional two-stage anaerobic digestion. However, further research is needed to confirm this hypothesis.

Credit author statement

Kavosh Karami: Methodology, Formal analysis, Resources, Data Curation, Writing- original draft preparation. **Safoora Mirmohamadsadeghi:** Data collection, analysis, and writing. **Keikhosro Karimi**: Methodology, Resources, Conceptualization, Resources, Supervision, Funding acquisition, Project administration, Writing-Review and Editing. **Rajeev Kumar:** Visualization, Supervision, Project administration, Writing- Review and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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